Customer Segmentation Using Unsupervised Learning

1. Project Overview

This project demonstrates the application of unsupervised machine learning techniques to customer behavioural data. The objective was to identify groups of customers with similar behavioural characteristics, detect potentially unusual customer records, and reduce the dimensionality of the dataset for visualization.

The project implements four major machine learning techniques:

- K-Means clustering
- DBSCAN clustering
- Isolation Forest anomaly detection
- Principal Component Analysis (PCA)

The implementation uses Python and Scikit-learn, with Pandas and NumPy for data manipulation and Matplotlib for visualization.

2. Problem Statement

Businesses often collect customer information such as monthly spending, transaction frequency, and account age. However, raw customer records do not automatically reveal meaningful customer segments. The goal of this project was to explore whether customers could be grouped based on behavioural similarities without providing predefined customer categories or labels. This is an unsupervised learning problem because the dataset does not contain target labels that tell the algorithms which customer belongs to which group.

3. Dataset Description

A synthetic dataset containing 600 customer records was generated. Each customer record contains the following features:

Feature	Description
customer_id	Unique customer identifier
monthly_spending	Customer's monthly spending
transactions_per_month	Number of monthly transactions
account_age_months	Length of time the customer has maintained an account

The synthetic dataset was generated using three underlying customer profiles:

- **Premium customers:** higher monthly spending, more transactions, and longer account history.
- **Regular customers:** moderate spending, transaction frequency, and account age.
- **Occasional customers:** lower spending, fewer transactions, and shorter account history.

Each profile contributed 200 records, giving a total of 600 customers. These profiles were used to create the synthetic data. They were not supplied as labels to the clustering algorithms.

4. Data Preprocessing

Before applying the clustering algorithms, the dataset was loaded and its machine learning features were selected. The customer_id column was excluded because it is an identifier rather than a meaningful behavioural measurement. The following features were used:

- monthly_spending
- transactions_per_month
- account_age_months

Feature Scaling

StandardScaler was used to standardize the numerical features. This was necessary because monthly spending has a substantially larger numerical scale than transaction frequency and account age. Without scaling, spending could disproportionately influence distance-based algorithms such as K-Means and DBSCAN. The resulting standardized features were used for the experiments.

5. K-Means Clustering

K-Means was used to divide the customers into three clusters. The model was configured with:

- Number of clusters: 3
- Random state: 42
- Number of initializations: 10

Results

Silhouette Score: 0.6661

Cluster distribution:

Cluster	Number of Customers
Cluster 0	200
Cluster 1	200
Cluster 2	200
Total	**600**

Interpretation

The silhouette score of 0.6661 indicates reasonably strong separation between the discovered clusters. The equal distribution of 200 customers per cluster is consistent with the balanced synthetic dataset. However, cluster numbers are arbitrary identifiers. Further analysis of cluster feature averages is required before assigning business names such as Premium, Regular, or Occasional.

6. Selecting the Number of Clusters

The number of clusters was evaluated using two metrics: Inertia and Silhouette Score. The experiment evaluated cluster counts from 2 through 7.

Results

K	Inertia	Silhouette Score
2	434.17	0.6609
3	184.03	0.6661
4	141.81	0.5644
5	126.44	0.5507
6	113.09	0.4287
7	104.71	0.4245

Interpretation

Inertia decreased as the number of clusters increased. This is expected because increasing the number of centroids generally reduces the distance between observations and their assigned cluster centres. The silhouette score reached its highest value at K = 3. This suggests that three clusters provide the strongest separation among the tested configurations. The elbow curve and silhouette score therefore support using three clusters for this synthetic dataset.

7. DBSCAN Clustering

DBSCAN was implemented to explore density-based clustering. Unlike K-Means, DBSCAN does not require the number of clusters to be specified in advance. The experiment used:

- Epsilon (eps): 0.8
- Minimum samples (min_samples): 5

Results

Metric	Result
Clusters discovered	2
Noise points	0
Total customers	600
Silhouette Score	0.6609

Cluster distribution:

Cluster	Customers
Cluster 0	200
Cluster 1	400

Interpretation

DBSCAN identified two density-based clusters and classified no observations as noise. This demonstrates that DBSCAN can discover a different grouping structure from K-Means, even when applied to the same standardized dataset. The result also shows that DBSCAN's output depends on its density parameters, particularly epsilon and minimum samples. Further experiments with different epsilon values would help evaluate how sensitive the discovered clusters are to these parameters.

8. Anomaly Detection Using Isolation Forest

Isolation Forest was used to identify observations with unusual combinations of customer features. The model was configured with:

- Contamination: 0.05
- Number of estimators: 100
- Random state: 42

Results

Classification	Customers
Normal	570
Anomalous	30
Total	**600**

The model classified 30 customers as anomalous, representing 5% of the dataset.

Interpretation

Isolation Forest identifies observations that are relatively unusual compared with the overall data distribution. The 5% contamination parameter controls the expected proportion of observations treated as anomalies for this experiment. These records should not automatically be interpreted as fraudulent or invalid customers. They are statistical anomalies that would require further investigation and business context.

9. Dimensionality Reduction Using PCA

Principal Component Analysis (PCA) was applied to reduce the three standardized customer features to two principal components. The purpose was to retain as much information as possible while making the dataset easier to visualize.

Results

Principal Component	Explained Variance
PC1	92.89%
PC2	4.36%
Total	**97.25%**

Interpretation

The first two principal components preserve 97.25% of the variance in the original three-dimensional dataset. This means that the two-dimensional representation retains most of the variation present in the original features. PCA therefore provides a useful lower-dimensional representation for visual inspection of customer patterns. PCA does not itself perform clustering. In this project, K-Means cluster labels were used to colour the PCA visualization.

10. Project Architecture

The project was organized into separate Python modules to keep data generation, preprocessing, modelling, and visualization responsibilities independent.

```
customer-segmentation-ml/
├── data/
│   ├── customers.csv
│   ├── kmeans_results.csv
│   ├── dbscan_results.csv
│   ├── anomaly_detection_results.csv
│   └── cluster_comparison_results.csv
├── models/
├── plots/
│   ├── cluster_comparison.png
│   └── pca_visualization.png
├── src/
│   ├── generate_data.py
│   ├── preprocessing.py
│   ├── clustering.py
│   ├── cluster_comparison.py
│   ├── dbscan_experiment.py
│   ├── anomaly_detection.py
│   └── dimensionality_reduction.py
├── main.py
├── requirements.txt
└── README.md
```

The `main.py` file coordinates the complete workflow and executes the experiments.

11. Technologies Used

- Python
- NumPy
- Pandas
- Scikit-learn
- Matplotlib

12. Key Findings

The experiments produced the following findings:

1. K-Means achieved its highest silhouette score of 0.6661 with three clusters.
2. The three K-Means clusters each contained 200 customers.
3. DBSCAN discovered two clusters and identified no noise points using the selected parameters.
4. Isolation Forest identified 30 potentially anomalous customer records.
5. PCA retained 97.25% of the original dataset's variance using two principal components.

Together, these experiments demonstrate how different unsupervised learning techniques can reveal different structures and patterns in customer data.

13. Limitations and Future Improvements

This project uses synthetic data with three deliberately generated customer profiles. Therefore, its results should not be treated as proof that real-world customer populations naturally divide into three equally sized groups.

Potential improvements include:

- Test the clustering algorithms against real-world customer datasets.
- Experiment with different DBSCAN epsilon and minimum-sample values.
- Analyse the average feature values of each discovered cluster.
- Investigate anomalous records using domain-specific rules.
- Persist trained models and preprocessing objects.
- Add automated tests and model evaluation reports.
- Explore clustering performance on larger datasets.

14. Conclusion

This project provided practical experience implementing and evaluating unsupervised machine learning techniques. It demonstrated how clustering algorithms can discover patterns without predefined labels, how anomaly detection can identify unusual observations, and how dimensionality reduction can simplify high-dimensional data for visualization.

The project establishes a practical foundation for understanding how unsupervised machine learning can support data exploration, customer analytics, and intelligent software systems.